Rajat Varma

Senior Software Engineer

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WORK EXPERIENCE

Senior Software Engineer | Microsoft *(Jan 2026 - Present)*
Software Engineer | Microsoft *(Mar 2025 - Jan 2026)*

- Lead the design of a microservice for agent orchestration at scale, with observability built into its operating model.
- Developed and optimized RAG and context pipelines to improve agent performance and efficiency.
- Built a production search agent; collaborate across teams to improve performance, reliability and user experience.

Software Development Engineer II | Flipkart *(Jun 2023 - Mar 2025)*
Software Development Engineer | Flipkart *(Jun 2022 - Jun 2023)*

- Improved service reliability for Big Billion Days peak traffic, helping sustain events serving millions of requests.
- Maintained and optimized a payment backend under strict latency and correctness constraints.
- Built a RAG pipeline for user data and metrics, improving data observability and investigation of production behavior.

PROJECTS

IncidentLab - AI reliability | **Source Code** | **Live demo**

- Built an incident investigation and repair-verification system with cited evidence, human approval and fail-closed policy checks.
- Implemented an independent verifier in a network-disabled Docker sandbox using React, FastAPI, PostgreSQL, Temporal and OpenTelemetry.

Agent Workbench - Coding agent evaluation | **Source Code** | **Live demo**

- Built a local-first evaluation lab with frozen task suites, immutable configuration snapshots, isolated Git worktrees and independent graders.
- Retained traces and diffs to make regressions inspectable in an Electron, React, FastAPI and SQLite desktop app with Codex CLI.

Juniper - Voice AI | **Source Code** | **Live demo**

- Built a voice receptionist for booking and rescheduling with explicit confirmation, duplicate-safe retries and durable session recovery.
- Enforced database constraints against overlapping appointments with TypeScript, Python, LiveKit, Fastify and PostgreSQL.

LLM Gateway - LLM infrastructure | **Source Code** | **Case study**

- Built an OpenAI-compatible Go gateway with bounded streaming, primary/backup routing and shared Redis token admission.
- Implemented per-attempt usage accounting in SQLite, Prometheus metrics and Grafana dashboards for provider and request visibility.

SKILLS

Engineering: Python, Go, TypeScript, React, FastAPI
Databases & Messaging: PostgreSQL, SQLite, Redis, Kafka
AI Systems: RAG, agent orchestration, evaluations, LLMOps, sandboxing, guardrails
Infrastructure: Kubernetes, containers, orchestration, CI/CD, Prometheus, Grafana

EDUCATION

Degree / Examination	Institution / Board
BTech, Computer Science & Engineering	Ramaiah Institute of Technology, Bengaluru